Instruction :
Prepare a test strategy document to test applications on Wealth Banking domain
-Ask me 5 questions one by one for Test Strategy creation
-Ask me the next quesion after me providing the answer
Step by Step implementation
Context :
As a tester who has sound knowledge in Wealth Banking domain to create test strategy

# WEALTH BANKING
## TEST STRATEGY DOCUMENT

### Enterprise Investment Platform
Major Version Upgrade / Migration

| | |
|---|---|
| **Document Version** | 1.0 |
| **Classification** | Confidential |
| **Prepared By** | QA / Test Strategy Team |
| **Date** | 06 April 2026 |
| **Status** | Draft for Review |

# 1. Introduction

This Test Strategy document defines the approach, scope, objectives, and methodology for quality assurance of the Wealth Banking Enterprise Platform undergoing a Major Version Upgrade and Migration. It serves as the governing framework for all testing activities across the engagement lifecycle.

The platform under test encompasses six core modules: Investment Advisory, Portfolio Management, Client Onboarding & KYC, Risk & Compliance Management, Trading & Order Management, and Private Banking CRM. These components are tightly integrated and serve High-Net-Worth (HNW) and Ultra-High-Net-Worth (UHNW) client segments, demanding the highest standards of accuracy, security, and regulatory adherence.

# 2. Test Objectives

The primary objectives of this test strategy are:

- Validate the functional correctness of all six platform modules post migration, ensuring business logic parity with the legacy system.
- Ensure data integrity and completeness across all migrated client portfolios, transactions, and historical records.
- Verify seamless API and third-party integration with market data feeds, custodians, and regulatory reporting systems.
- Assess system resilience and performance under peak load conditions consistent with trading hours and end-of-day batch processing.
- Identify and eliminate security vulnerabilities through structured penetration and security testing.
- Achieve all defined exit criteria prior to production go-live.

# 3. Scope of Testing

## 3.1 In Scope

| Module | Key Testing Areas |
|---|---|
| **Investment Advisory Platform** | Suitability assessments, model portfolio mapping, rebalancing triggers, MiFID advisory workflows, client risk profiling |
| **Portfolio Management System** | Holdings valuation, asset allocation, P&L computation, corporate actions, performance attribution |
| **Client Onboarding / KYC** | Identity verification, document upload & validation, AML screening, risk categorisation, onboarding workflow state machine |
| **Risk & Compliance Management** | Limit breach alerting, exposure calculations, regulatory report generation, audit trail integrity |
| **Trading & Order Management** | Order lifecycle (placement to settlement), FIX protocol messaging, pre-trade checks, execution quality |
| **Private Banking CRM** | Client 360 view, relationship hierarchy, activity logging, pipeline management, notification workflows |

## 3.2 Out of Scope

- Infrastructure provisioning and on-premise server configuration (handled by DevOps/Infrastructure team).

- Third-party vendor black-box systems where API contracts are unchanged and regression-verified.
- Disaster recovery failover testing (separate DR test plan in progress).
- User training and change management activities.

# 4. Test Approach & Methodology

The test approach follows a risk-based, shift-left testing model integrated with the CI/CD pipeline, ensuring early defect detection and continuous feedback throughout the migration lifecycle.

## 4.1 Functional Testing

Functional testing covers end-to-end business workflows across all six modules. Test cases are derived from Business Requirement Specifications (BRS), functional design documents, and legacy system behaviour baselines. Coverage includes positive, negative, and boundary value scenarios.

- Investment Advisory: Suitability engine rule validation, model portfolio allocation accuracy, rebalancing threshold triggers.
- Portfolio Management: Mark-to-market calculations, multi-currency valuation, corporate action processing (dividends, splits, mergers).
- KYC Onboarding: Document workflow state transitions, AML watchlist screening, risk tier classification logic.
- Trading & OMS: Full order lifecycle from placement through confirmation and settlement; FIX message integrity.
- Risk & Compliance: Breach alert accuracy, limit utilisation calculations, regulatory report field-level validation.
- CRM: Client hierarchy relationships, activity association, pipeline stage transitions, notification delivery.

## 4.2 Regression Testing

A comprehensive regression suite will be maintained to validate that migration changes do not break existing functionality. Regression packs will be executed at every sprint release and at major milestones, leveraging Selenium-based automated UI tests and REST Assured API test suites.

- Automated regression coverage target: ≥ 80% of critical business workflows.
- Regression suite will be integrated into the Jenkins CI/CD pipeline for nightly execution.
- Delta regression analysis will identify impacted test areas for each deployment build.

## 4.3 API & Integration Testing

API testing will be conducted using REST Assured and Postman to validate all internal microservice interfaces and external third-party integrations.

- Validate request/response schemas, status codes, error handling, and data transformation accuracy.
- Integration points: Market data providers (Bloomberg / Refinitiv), custodian platforms, core banking ledger, regulatory reporting engines.
- Contract testing will be performed to ensure backward-compatible API versioning post migration.
- End-to-end integration chain testing for critical flows: Order placement → Execution → Portfolio update → Client Statement.

## 4.4 Security & Penetration Testing

Security testing will be conducted in a dedicated on-premise test environment mirroring production configurations. The scope covers application-layer and data-layer security.

- OWASP Top 10 vulnerability assessment for all web-facing application components.

- Role-based access control (RBAC) validation: Relationship Manager, Compliance Officer, Portfolio Manager, Administrator roles.
- Sensitive data masking verification for PII, account numbers, and transaction values in logs and reports.
- Session management and authentication testing including MFA, token expiry, and concurrent session controls.
- SQL injection, XSS, CSRF, and broken object-level authorisation (BOLA) checks on all API endpoints.

### 4.5 Performance & Load Testing

Performance testing will be executed using JMeter to simulate production-equivalent load and validate system behaviour under stress. Benchmarks are defined based on SLA agreements.

| Test Scenario | Target SLA | Acceptance Threshold |
|---|---|---|
| Portfolio valuation (peak load) | < 3 sec response | 95th percentile ≤ 3s at 500 concurrent users |
| Order placement (trading hours) | < 1 sec response | 99th percentile ≤ 1s at 300 concurrent users |
| KYC document submission | < 5 sec end-to-end | Error rate < 0.1% under sustained load |
| End-of-day batch processing | Complete within 2 hrs | Full portfolio recalculation for 50,000 accounts |
| Report generation | < 10 sec per report | 95th percentile ≤ 10s for complex multi-asset reports |

## 5. Test Environment Strategy

All test environments are hosted on-premise, isolated from production infrastructure. Environment access is governed by strict RBAC and network segmentation policies consistent with the Wealth Banking security framework.

| Environment | Purpose | Configuration |
|---|---|---|
| **DEV** | Unit & Component | Developer sandbox, lightweight data, daily refresh |
| **SIT** | System Integration Testing | Full application stack, synthetic data, API integration verification |
| **UAT** | User Acceptance Testing | Production-equivalent, anonymised production data subset |
| **PERF** | Performance & Load Testing | Scaled production topology, JMeter load injectors, APM monitoring |
| **SEC** | Security Testing | Isolated environment for penetration testing, vulnerability scanning |

## 6. Tools & Automation Framework

| Category | Tool | Usage |
|---|---|---|

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|---|---|---|
| **UI Automation** | Selenium WebDriver | Cross-browser regression, workflow automation for all six modules |
| **Mobile Testing** | Appium | Mobile app functional and regression testing for wealth client app |
| **API Testing** | REST Assured / Postman | API contract, functional, and integration test automation |
| **Performance Testing** | Apache JMeter | Load, stress, soak, and spike testing with real-time dashboards |
| **CI/CD Integration** | Jenkins / Azure DevOps | Pipeline-triggered test execution, build gate quality checks |
| **Defect Management** | JIRA | Defect logging, triage, severity tracking, and resolution workflow |
| **Test Management** | Zephyr / TestRail | Test case repository, execution tracking, and reporting |
| **Code Repository** | Git / Bitbucket | Version control for automation scripts and test data |

# 7. Defect Management

## 7.1 Defect Severity Classification

| Severity | SLA (Fix) | Description |
|---|---|---|
| **Critical (S1)** | Same business day | System crash, data loss, incorrect financial calculations, security breach, trade execution failure |
| **High (S2)** | Within 2 business days | Core workflow blocked, regulatory report inaccuracy, KYC screening failure |
| **Medium (S3)** | Within 5 business days | Non-critical feature malfunction, UI/UX deviation, non-blocking workflow issue |
| **Low (S4)** | Next release cycle | Cosmetic defects, minor UI alignment, enhancement suggestions |

## 7.2 Defect Lifecycle

New → Open → In Progress → Fixed → Re-test → Closed / Rejected. All Critical and High defects require root cause analysis (RCA) documentation before closure. Defect leakage metrics will be tracked and reported weekly to the Project Steering Committee.

# 8. Risks & Mitigations

| Risk | Impact | Mitigation |
|---|---|---|
| Incomplete data migration from legacy system | **High** | Pre-migration data profiling; automated data reconciliation scripts; dual-run parallel validation |
| API contract changes breaking downstream integrations | **High** | Consumer-driven contract testing with Pact; API versioning governance; integration regression suite |

| | | |
|---|---|---|
| Environment instability impacting test execution | **Medium** | Daily environment health checks; backup environment provisioning; rollback procedures documented |
| Security vulnerabilities in migrated codebases | **High** | SAST/DAST tooling integrated in CI pipeline; dedicated penetration test cycle before UAT |
| Scope creep from new regulatory requirements mid-migration | **Medium** | Change control board governance; impact analysis mandatory for all regulatory change requests |
| Performance degradation under trading peak hours | **High** | Capacity planning review; JMeter load tests executed in PERF environment with production data volumes |
| Third-party market data feed unavailability in test environments | **Medium** | Data virtualisation / mock services for market feeds; vendor coordination for test environment access |

# 9. Entry & Exit Criteria

## 9.1 Entry Criteria

- Test environments provisioned, configured, and verified as stable.
- Test data sets prepared and loaded, including anonymised production data subsets.
- All test cases reviewed, approved, and baselined in the test management tool.
- Application build deployed to SIT/UAT with release notes and known issues documented.
- Automation framework and CI/CD pipeline integration verified and operational.

## 9.2 Exit Criteria (Go-Live Gate)

| Exit Criterion | Target | Measurement |
|---|---|---|
| Test case pass rate | **≥ 95%** | Executed vs Passed in test management tool |
| Zero Critical (S1) or High (S2) open defects | **100% resolved** | JIRA defect dashboard at release gate |
| All regulatory test cases passed | **100%** | Dedicated regulatory test suite execution report |
| Performance SLAs met | **All benchmarks achieved** | JMeter test reports, APM dashboards |
| Signed UAT sign-off obtained | **All stakeholders signed** | UAT acceptance form from Business and Compliance owners |
| Security penetration test report reviewed | **No critical findings open** | Pen test report accepted by CISO |
| Data migration reconciliation report approved | **Zero unreconciled records** | Automated reconciliation report sign-off |

# 10. Roles & Responsibilities

| Role | Responsibilities |
|---|---|
| **Test Manager** | Test strategy ownership, resource planning, stakeholder reporting, risk escalation, exit gate governance |

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|---|---|
| **Lead QA Analyst (Wealth Domain)** | Test case design, domain-specific review, defect triage leadership, UAT coordination |
| **Automation Engineer** | Selenium and REST Assured framework development, CI/CD integration, regression suite maintenance |
| **Performance Test Engineer** | JMeter script development, load profile design, performance report analysis and recommendations |
| **Security Test Analyst** | Penetration testing, OWASP assessment, security report preparation and finding remediation tracking |
| **Business Analyst** | Requirements clarification, acceptance criteria definition, UAT participation |
| **DevOps / Environment Manager** | Environment provisioning, deployment coordination, build management |

## 11. Test Deliverables

| Deliverable | Owner | Milestone |
|---|---|---|
| Test Strategy Document (this document) | Test Manager | **Sprint 1** |
| Master Test Plan | Test Manager | **Sprint 1** |
| Functional Test Cases (all 6 modules) | Lead QA Analyst | **Sprint 2** |
| Automation Framework & Regression Suite | Automation Engineer | **Sprint 3** |
| Data Migration Test Report | Lead QA Analyst | **Post-migration run** |
| API Integration Test Report | Automation Engineer | **SIT completion** |
| Performance Test Report & Benchmark Results | Perf Test Engineer | **Pre-UAT** |
| Security Penetration Test Report | Security Analyst | **Pre-UAT** |
| UAT Test Execution Report | Test Manager | **UAT completion** |
| Go / No-Go Test Summary Report | Test Manager | **Release gate** |

## 12. Document Approval & Sign-Off

This Test Strategy document requires formal review and sign-off from the following stakeholders before testing activities commence. Sign-off indicates agreement with the defined approach, scope, and exit criteria.

| Name | Role | Signature | Date |
|---|---|---|---|
| | Test Manager | | |
| | Head of Wealth Technology | | |
| | Chief Compliance Officer | | |

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|---|---|---|---|
| | Business Sponsor | | |

*— End of Test Strategy Document —*